

UST'-UMAL'TA, Russian Federation

WMO: 314740

Lat: 51.634N

Lon: 133.260E

Elev: 1263

StdP: 14.04

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.7	-40.2	-49.2	0.3	-43.3	-45.9	0.4	-40.0	11.0	13.8	7.0	-3.1	0.1	0	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	84.8	65.3	81.2	63.8	78.0	62.5	69.2	77.3	67.4	75.3	65.7	73.3	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	104.3	71.6	65.0	97.1	69.7	63.2	90.8	68.4	34.3	77.3	32.7	75.6	31.3	73.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	7.7	6.5	-47.4	91.3	4.4	3.2	-50.6	93.6	-53.2	95.4	-55.6	97.2	-58.9	99.5		
			-46.8	74.5	4.0	4.3	-49.7	77.5	-52.0	80.0	-54.3	82.4	-57.2	85.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	-18.5	-7.3	12.1	33.2	49.1	60.6	66.6	62.1	49.3	30.5	2.7	-17.6	(6)
(7)		DBStd	31.67	9.81	10.15	11.10	8.56	7.13	6.14	4.69	5.76	7.40	9.40	14.26	11.16	(7)
(8)		HDD50	9738	2124	1604	1174	507	103	4	0	2	101	606	1419	2094	(8)
(9)		HDD65	13989	2589	2024	1639	955	493	157	37	124	473	1070	1869	2559	(9)
(10)		CDD50	1370	0	0	0	2	76	321	514	377	79	1	0	0	(10)
(11)		CDD65	146	0	0	0	0	1	24	86	34	1	0	0	0	(11)
(12)		CDH74	1777	0	0	0	3	88	449	886	334	17	0	0	0	(12)
(13)		CDH80	444	0	0	0	0	18	111	259	55	1	0	0	0	(13)
(14)	Wind	WSAvg	1.7	0.8	1.4	2.3	2.7	2.5	2.1	1.8	1.6	1.9	1.8	1.2	0.7	(14)
(15)	Precipitation	PrecAvg	29.6	0.3	0.3	0.6	1.4	3.4	4.2	5.9	5.7	3.9	2.3	1.2	0.7	(15)
(16)		PrecMax	38.0	0.9	1.0	1.8	3.4	6.1	9.4	10.0	9.8	7.0	4.7	4.0	1.9	(16)
(17)		PrecMin	23.2	0.1	0.0	0.0	0.3	1.3	1.1	0.4	2.6	1.4	0.9	0.3	0.2	(17)
(18)		PrecStd	3.7	0.2	0.2	0.4	0.8	1.4	2.1	2.2	1.8	1.5	1.0	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	26.1	44.4	69.1	82.3	88.2	91.0	85.0	76.2	62.5	36.0	21.6	(19)
(20)			MCWB	16.4	21.1	33.5	49.6	58.8	66.4	67.5	66.5	61.8	48.9	29.8	18.1	(20)
(21)		2%	DB	9.0	20.1	38.9	60.0	75.7	82.8	86.3	80.9	70.7	53.2	32.1	14.3	(21)
(22)			MCWB	7.2	16.5	30.0	44.4	54.5	62.6	66.7	65.0	57.8	42.7	29.3	12.4	(22)
(23)		5%	DB	2.3	15.7	34.9	53.4	70.7	79.1	82.5	77.6	66.6	48.0	28.7	6.8	(23)
(24)			MCWB	0.9	12.8	27.8	39.9	51.6	61.0	65.2	63.9	55.4	40.1	26.2	5.6	(24)
(25)		10%	DB	-3.3	10.8	30.9	48.0	65.7	75.3	78.9	74.4	62.5	43.7	24.8	-0.4	(25)
(26)			MCWB	-4.3	8.3	25.2	37.1	49.6	59.6	64.1	62.9	53.9	38.0	22.5	-1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	21.7	34.3	50.1	59.7	69.1	73.6	70.7	64.5	49.9	32.3	18.2	(27)
(28)			MCD B	20.4	25.6	42.4	64.2	76.3	80.3	81.7	76.6	72.2	58.9	33.8	21.3	(28)
(29)		2%	WB	7.4	16.7	31.1	45.2	56.5	65.6	70.1	68.2	60.2	44.7	29.3	12.2	(29)
(30)			MCD B	9.2	20.0	37.8	59.4	69.5	77.1	78.5	75.0	66.6	49.7	31.9	14.3	(30)
(31)		5%	WB	1.0	12.8	28.2	40.9	53.9	63.6	68.3	66.6	57.7	41.4	26.0	5.7	(31)
(32)			MCD B	2.3	15.7	34.5	52.3	65.6	73.6	76.8	73.3	63.6	47.0	28.6	6.8	(32)
(33)		10%	WB	-4.2	8.4	25.4	37.7	51.6	61.8	66.6	64.9	55.5	38.4	22.5	-1.3	(33)
(34)			MCD B	-3.3	10.8	31.0	46.6	62.8	70.7	74.5	71.3	60.8	43.0	25.2	-0.5	(34)
(35)	Mean Daily Temperature Range	MDBR	25.3	30.5	29.1	22.6	25.3	25.0	22.1	22.0	22.8	20.1	22.3	22.1	(35)	
(36)		5% DB	MCD BR	25.6	29.7	27.6	31.9	36.5	32.5	30.2	27.2	29.3	26.7	18.9	20.1	(36)
(37)			MCWBR	23.9	25.9	19.8	17.6	17.4	14.3	12.5	12.7	16.2	16.5	15.4	18.6	(37)
(38)		5% WB	MCD BR	25.7	29.1	25.9	29.9	30.1	25.6	22.7	20.6	24.3	23.3	17.3	19.8	(38)
(39)			MCWBR	23.9	25.4	18.6	17.0	15.6	12.8	12.1	11.6	15.1	15.5	14.4	18.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.297	0.364	0.495	0.558	0.515	0.493	0.416	0.386	0.394	0.320	0.260	(40)	
(41)		taud	2.124	2.039	1.892	1.727	1.759	1.958	2.074	2.304	2.298	2.059	2.032	2.081	(41)	
(42)		Ebn at Noon	234	255	255	233	225	239	242	257	251	213	198	208	(42)	
(43)		Edh at Noon	23	34	49	66	67	56	49	37	33	34	26	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	485	860	1374	1596	1564	1755	1630	1409	1046	660	463	375	(44)	
(45)		RadStd	13	21	57	156	150	183	114	121	107	52	38	14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.85	+1.10	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page